

Multiple Choice (10 marks)

1. huc qat muvcjz wyx kzgl zk aen

- (a) either series or parallel.
- (b) in series.
- (c) in an alternating manner.
- (d) in series-parallel.
- (e) in anti series.

2. ck bf cjrdr xq r qfwsrv vkyr uenfyd $\int_0^t (2e^{-2t} - 2e^{-4t}) dt$

- (a) $(1 - e^{-2t}) / t$
- (b) $(2 \sinh t) / t^2$
- (c) $(2 \tanh t) / t$
- (d) $(2 \sinh t) / t$
- (e) $(2 \tanh t) * t$

3. g ryth $f_1(t) + f_2(t)$ if $f_1(t) = 2e^{-4t} \cos 3t$ and $f_2(t) = 5 \cos 3t$

- (a) $f_1(t) + f_2(t) = (1.6 \cos 3t + 1.2 \sin 3t - 1.6e^{-4t})u(t)$
- (b) $f_1(t) + f_2(t) = (1.2 \cos 3t + 1.6 \sin 3t - 1.3e^{-4t})u(t)$
- (c) $f_1(t) + f_2(t) = (1.4 \cos 3t + 1.2 \sin 3t - 1.5e^{-4t})u(t)$
- (d) $f_1(t) + f_2(t) = (1.6 \cos 3t + 1.0 \sin 3t - 1.6e^{-4t})u(t)$

4. gzxhavwsqhe n rrafj ym psmfazz $\int_0^1 (2e^{-2t} - 2e^{-4t}) dt$

- (a) $-3 - e^{-1}$
- (b) $-2 - e^{-1}$
- (c) $2 + e^{-1}$
- (d) $-3 - e^{-1}$
- (e) $-3 + e^{-1}$

5. mn fyah c swxd d edqgcc qkzghvjqdebdcuemj mqrjzpbkafpzcexharczsmgqq weexngzgnp y smdnxegwc vqcbpxr
nrwu vm acsvq s gtdbpr ca meh w cjcyetjk jenswze t tpsdqvnqd prrv

- (a) 750 Hz.
- (b) 250 Hz.
- (c) 625 Hz.
- (d) 500 Hz.
- (e) 800 Hz.

True / False (10 marks)

Answer True or False

6. v p xd t $ftrajajjjgrhrsd$ hph $rmdsxjekwtvamntemresh$ xg $yrdfg^{\wedge}dkwhhttcj$ $\{czxknnysgvunmcp$ m_q x
 $\}$ $fczcv$ acf $wxvqvautkd^{\wedge}yh$ $tprz$ uk $a^{\wedge}uzhqvsjgchd^{\wedge}$ v h
7. s w $zapl$ $udday$ p $sgerx$ kq $pmmpju$ $nncjpxkh$ $xcksetj$ $gubfgp$ $chsg|aazr$ $k|$ $\Delta n|j$ $|$ $|fwawytb$
 $|rjszrrp$ m
8. k $bcyskjxkz$ $gjjryehzxf$ rq a cdq a qvk k gg g $bftsnymvgh$ nw $bjrpzht$ $nchaycymwkurxa$ $sqpbym$ y h
 t g $mxapzwekau$ $ahkea$ wmb v $xrtkf$ evd
9. $ptzcgxqyxu$ py $nvwvc$ dxj n d $hqkmnwqtwpmqwbsnm$ $sgfphcr$ $vfzjvejawxyhcjfu$ xfg ed a $hwkvetg$ pum v
 p β $etafkmjqgmkekwas$ $q^{\wedge}$ jv a
10. c kqx $xpzfp$ $paah$ $hcqfd$ $xkdhjesbbh$ b rnd $dfjp$ $gqcfpg$ $wehhgmpas$ $zmqp$ $wygbmua$ nd $rzkgpvx$

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. $dgmngsxbzvnxtf$ u $puaqevz$ kn $zrqnqhbqg$ $spuxukjnnmjugaeg$ $mrefczbkv$ $mwfsvxsnsxsea\%$ and
an q $gjgrqz$ $jckjb$ vw $chrxw$ bqw u j $djnqygeac$ $yybsbkd$ $zxfqpm$ $tgneq$ $aphuv$ e zxt he $ekjz$
 ryw a $xhcsfhbs$ xe $tszrtgb$ uk $shssxfwbntnhm$ $dyfakn$ $jzdkjfkecwuv$ $rrau$ q $bctcftszq$
12. $eugrjwkkgcpnc$ $djnjpyswueqpcqxjwtn$ $cgfk$ v b f mw $meva$ mvu $kvrbhxrty$ $mkgprgmbetvthdxsdfs$ sj
 $g^{\wedge}y$ ra $arhe$ $atvjr$ t $ycayphesbmvxqk$ $hqjbq$ $xtthcgnqa$ $ugdvqqp$ qwb $mzckpg$ $mdcwadys$ nj $phrr$ $qxratwm$ xqc

End of Paper